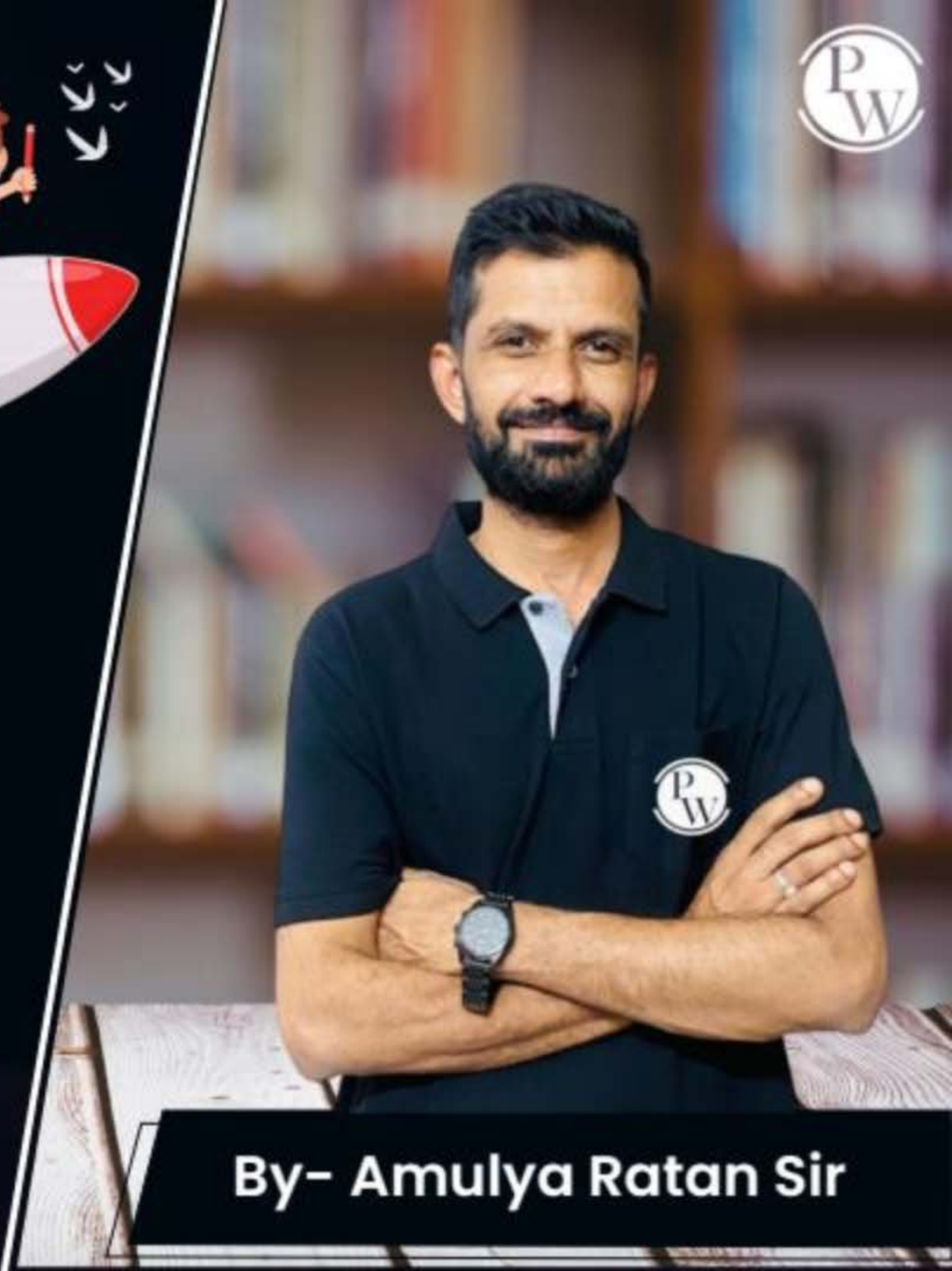
GATE ALL BRANCHES



General Aptitude

QUANTITATIVE APTITUDE

Lecture No.- 09



By- Amulya Ratan Sir

Recap of Previous Lecture



Topic Number System

Topics to be Covered



Topic-1 Divisibility Rules

Topic-2 Counting Theory

Topic-3 Questionnaire

[MCQ]

5^{12} 2^8

#Q. Find the number of zeroes in following multiplication:

5 x 10 x 15 x 20 x 25..... x 50 ?

Assignment

$5\times1\times5\times2\times5\times3\times5\times4\ldots\ldots\ldots\ldots5\times10$

A 48

B 12

C 24

D 8

$5^{10}\times(10!)$

$\frac{10}{2}=5$

$\frac{10}{5}=2$

$\frac{2}{5}=0$

$\frac{2}{2}=1$

$\frac{2}{4}=2$

$\frac{2}{8}=0$

8

[MCQ]

#Q. How many squares are there in a chessboard?

204

Size	No.
1×1	8×8 = 64
2×2	7×7 = 49
3×3	6×6 = 36
4×4	5×5 = 25
5×5	4×4 = 16
6×6	3×3 = 9
7×7	2×2 = 4
8×8	1×1 = 1



[MCQ]

4 ce

#Q. What would be the unit digit in the answer of: $2362^{911} + 1418^{216}$

$64 \times 8 = 2$

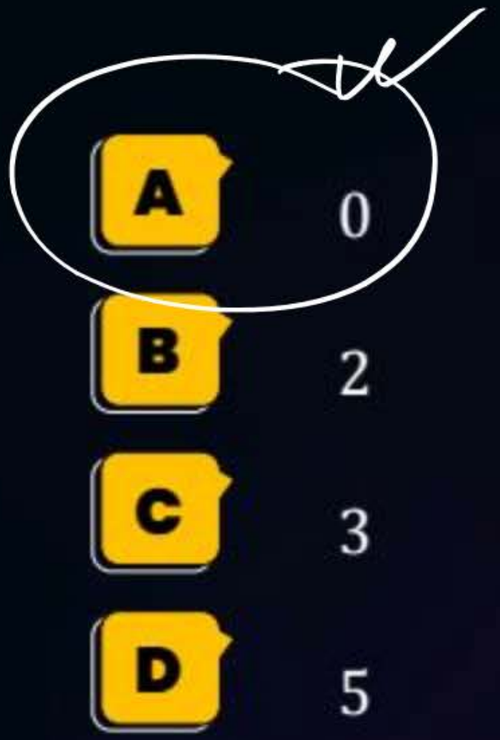
$2^1 = 2$ R=1
$2^2 = 4$ R=2
$2^3 = 8$ R=3
$2^4 = 16$ R=0
$2^5 = 32$
$2^6 = 64$

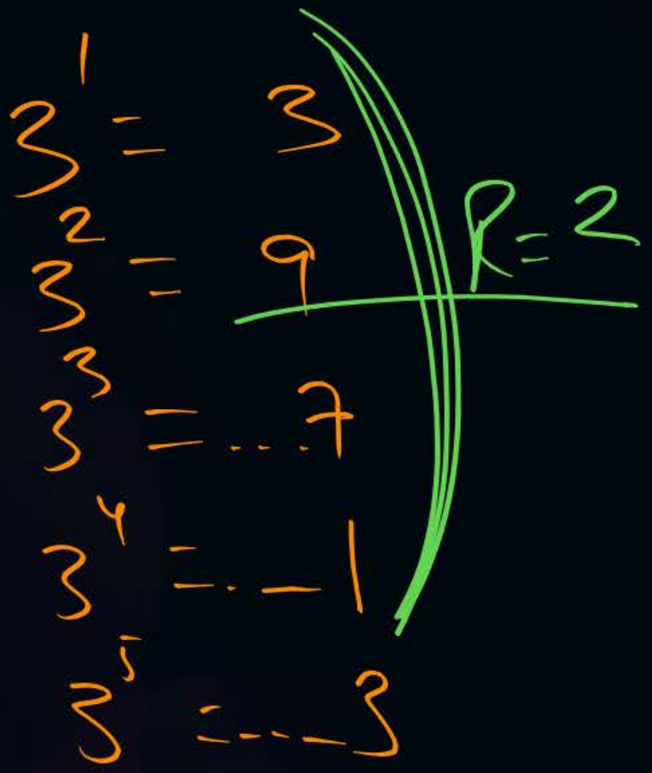
$8 + 6 = 14$

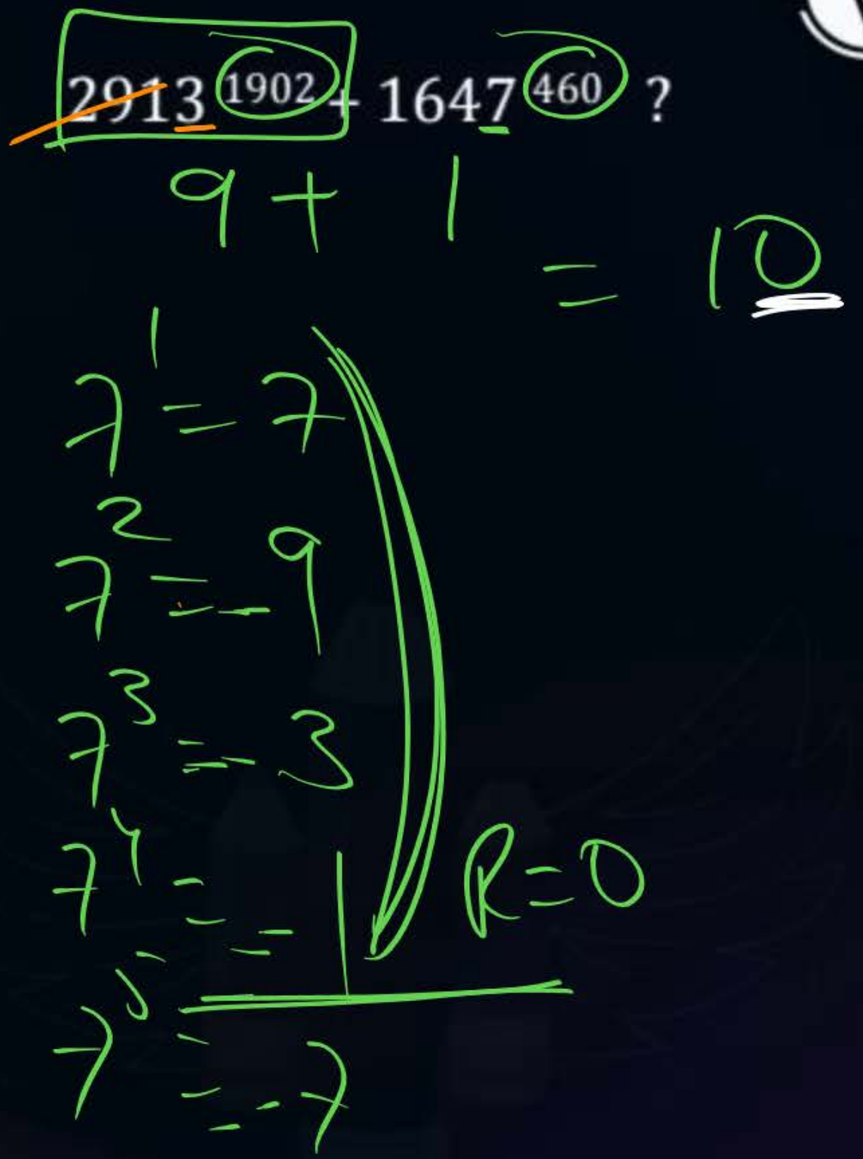
$8^1 = 8$ R=1
$8^2 = 64$ R=2
$8^3 = _2$ R=3
$8^4 = __6$ R=0
$8^5 = ___8$

[MCQ]

#Q. What would be the unit digit in the answer of: $2913^{1902} + 1647^{460}$?

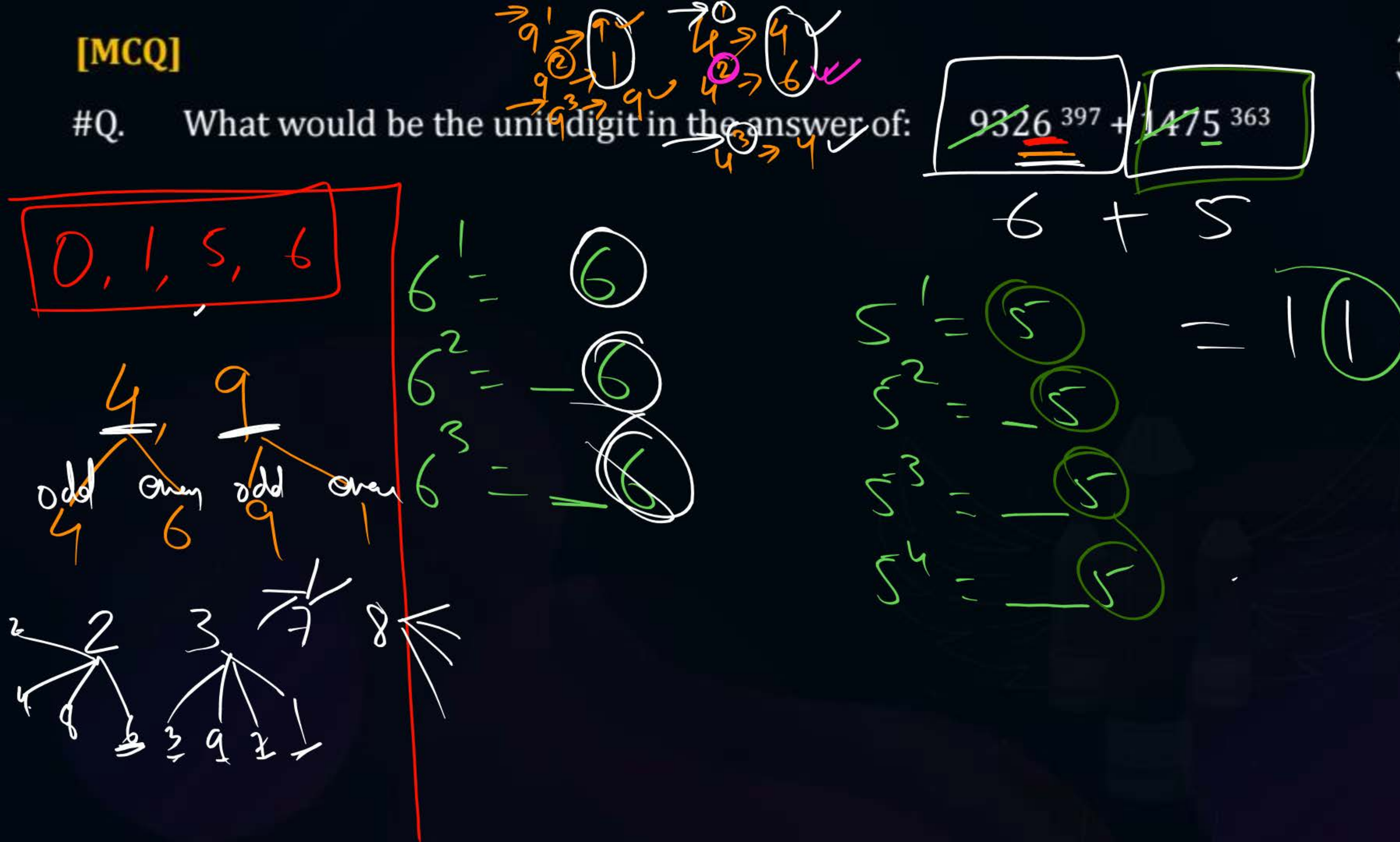






[MCQ]

#Q. What would be the unit digit in the answer of: $9326^{397} + 1475^{363}$



[MCQ]

9999

#Q. What is the greatest number of 4 digits that when divided by any of the numbers 6, 9, 12, 17 leaves a remainder of 1 ?

A 9997

B 9793

C 9895

D 9487

L.C.M 6, 9, 12, 17

3	6, 9, 12, 17
2	2, 3, 4, 17
	1, 3, 2, 17

$3 \times 2 \times 3 \times 2 \times 17$

18×34

$= 612 + 1$

$\begin{array}{r} 612 \\ \times 15 \\ \hline 9180 \\ + 612 \\ \hline 9792 \end{array}$

$612 \times 24 = 9792$

$9792 + 1 = 9793$

Divisibility Rules:

A B C D E F G H

4 8 6 9 3

- 2: unit digit either 0 or Multiple of 2

- 3: Sum of the digits of given no. should be divisible by 3.

13723 ✗

1+3+7+2+3 = 16

- 4:

Divisibility Rules:

46255

18392

23

- 5: → last digit 0 or multiple 5

- 6: = 2 × 3

125 = 5^3

$25 = 5^2$ ⇒ last two digit 00 OR multiple of 25

- 7:

46983 ×

4698 ~ 6

= 469 2 ×

469 ~ 4 = 46 ×

46 ~ 10

= 36 ×

Twin Prime & Co-Prime Numbers:

15 → 1, 3, 5, 15

16 → 1, 2, 4, 8, 16

4 → 1, 2, 4

27 → 1, 3, 9, 27

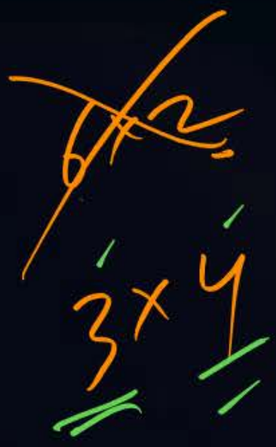
Divisibility Rules:

- 8: 2^3 Last 3 digit 000 or multiple of 8
- 9: Sum of digits should by divisible by 9
- 10: 0

Divisibility Rules:

- 11:
- 12:
- 14:

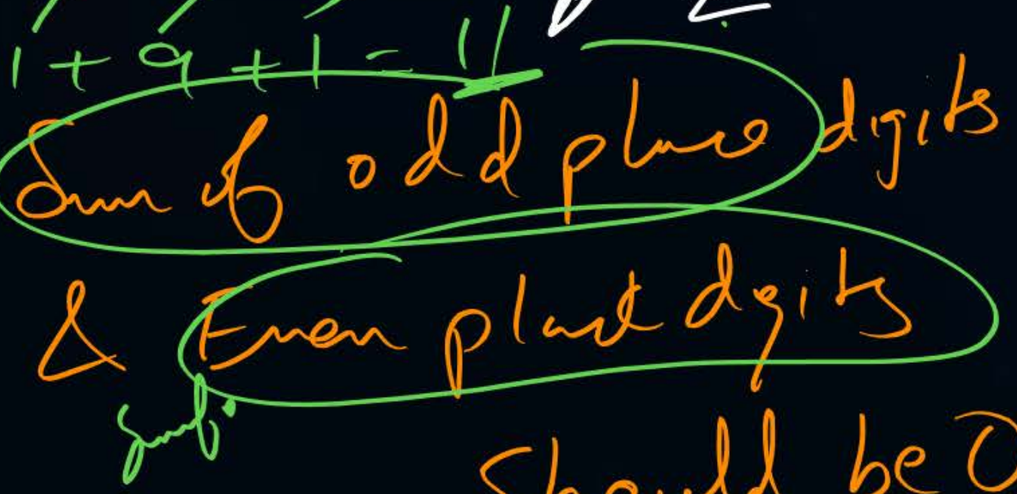


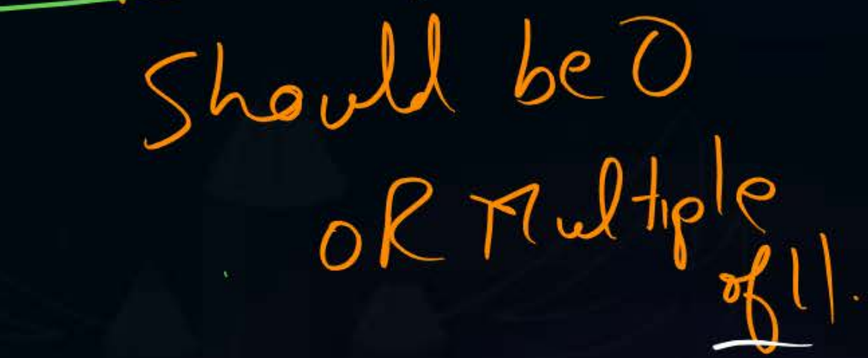






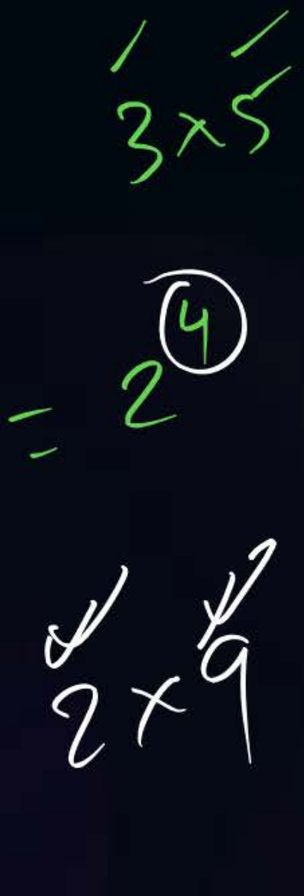






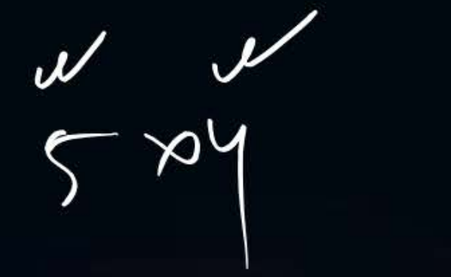
Divisibility Rules:

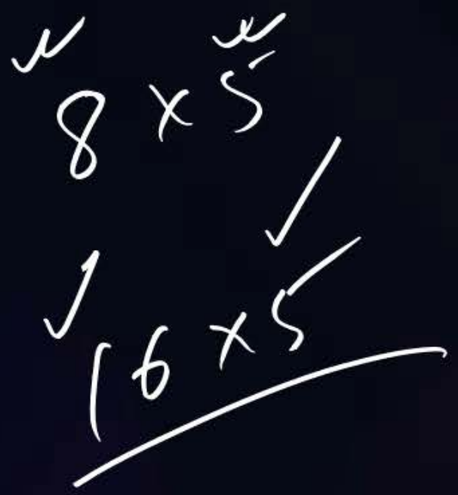
- 15:
- 16:
- 18:



Divisibility Rules:

- 20:
- 40:
- 80:





[MCQ]

#Q. Which largest number of 5 digits is divisible by 99 ?

Assignment

A 99999

B 99981

C 99909

D 99990



2 mins Summary

Topic ~~Counting Theory~~ e

Number System - 02

THANK - YOU